Comment on "Convergence Properties of the Likelihood of Computed Dynamic Models": A Corrected Proposition 2*

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Abstract

Proposition 2 of Fernández-Villaverde, Rubio-Ramírez, and Santos (2006) bounds the difference between the exact and the computed likelihood of a dynamic model by a term linear in the sample size. Ackerberg, Geweke, and Hahn (2009) showed by counterexample that the proposition is false; they did not locate the error in the proof or state a corrected result for the dynamic setting. We locate the error, a single step in which a difference of two T-fold products of densities is bounded as if the other factors were absent, and show that the proposition holds, with the same form of bound, for the logarithm of the likelihood: $|\log L_j(y^T) - \log L(y^T)| \leq (T\chi/\xi + K/(1-\alpha))\,\delta$ under a pathwise version of the paper's contraction condition. Linearity in the sample size T holds, but now in the log, rather than the level, of the likelihood. The average log likelihood is therefore within $O(\delta)$ of its exact value uniformly in the sample size, so the effect of a fixed approximation error δ on point estimates does not grow with T; relative to sampling uncertainty it matters once $\sqrt{T}\delta$ is not small. FRS's Bayesian corollaries hold with explicit, if conservative, constants.

KEYWORDS: Approximation error, likelihood, computed dynamic models, Markov chains, coupling.

1 Introduction

Likelihood-based estimation of a structural dynamic model uses an approximated policy function, so the researcher evaluates the likelihood L_j of the computed model rather than

*Every result in this comment has been verified in the Lean 4 proof assistant with the Mathlib library; Appendix D lists the correspondence, and the code is at `https://github.com/LeanEconomics/LeanEconomics`. The formal development and a first draft of this comment were produced with the assistance of Claude (Anthropic), an AI model.

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the likelihood L of the exact one. Fernández-Villaverde et al. (2006) (henceforth, FRS) asked how the two are related, in particular how numerical error incurred in solving the structural dynamic model affects likelihood-based estimation of that model. Their Proposition 1 shows that $L_j \to L$ for a fixed sample as the computed transition and measurement equations converge to the exact ones. Their Proposition 2 quantifies the speed: if the computed equations are within δ of the exact ones in the sup norm, then $|L_j(y^T) - L(y^T)| < (TB\chi + L_F/(1-\alpha))\delta$, with constants independent of the sample size T.[1] FRS drew three conclusions from it, emphasised in their abstract and introduction: that "second order approximation errors in the policy function ... have first order effects on the likelihood function"; that the error "gets compounded with the size of the sample", so the accuracy of the solution must improve as T grows if the computed likelihood is to stay close to the exact one; and hence that linearisation, which fixes δ, "is due to fail as the sample size grows". They also used the proposition to motivate a likelihood-ratio diagnostic for solution accuracy.

Ackerberg et al. (2009) (henceforth, AGH), following an unpublished note by Geweke (2007), showed that Proposition 2 is false. In the static Gaussian location model $y_t = \gamma + v_t$, approximated by $y_t = \gamma + \delta + v_t$, they computed a sharp bound on $|L_j - L|$ of order $(\sigma\sqrt{2\pi})^{-(T-1)}$. This grows exponentially in T when $\sigma < 1/\sqrt{2\pi}$ and vanishes exponentially when $\sigma > 1/\sqrt{2\pi}$, so FRS's linear bound is too small in one case and too large in the other. Their contribution on Proposition 2 is this counterexample, together with the argument that bounds on likelihood *levels* are the wrong object for inference, since in their example the effect of δ on the maximum likelihood estimator depends on neither T nor σ. They then proved new results for a static i.i.d. model, under a direct assumption that the log densities and their first two derivatives are within Δ_j of the exact ones: the pseudo-maximum-likelihood estimator has probability limit within $O(\Delta_j)$ of the truth, and is asymptotically normal if $\sqrt{T}\Delta_j \to 0$. AGH did not identify which step of FRS's proof fails and did not state a corrected version of Proposition 2; they remarked that seeking a valid bound on likelihoods in more general models "might not be a fruitful endeavor". Their constructive results are for the static model, and they take Δ_j as given rather than deriving it from the policy-function error δ and the dynamics of the state.

This comment locates the step of FRS's proof that fails and proves a corrected Proposition 2′ that keeps their intent. The correction is to bound the logarithm of the likelihood rather than its level. That is the scale on which likelihoods are used in estimation, and it turns FRS's bound on the absolute error into a bound on the relative error $|L_j/L - 1|$. The bound keeps FRS's form: under a pathwise version of their contraction condition, the log-likelihood error is at most $(T\chi/\xi + K/(1-\alpha))\delta$, linear in T, and this rate cannot be improved. FRS's three conclusions then hold in modified form. First, an approximation error of order δ in the policy function moves the log likelihood by $O(T\delta)$, but the average log likelihood by only $O(\delta)$, uniformly in T; this is AGH's "second order errors have second order effects", now derived in FRS's dynamic setting rather than assumed. Second, the effect

[1] Here L_F is a Lipschitz constant of an integrated conditional likelihood (Section 2). FRS denote it L, the same symbol as the likelihood; we write L_F throughout to avoid the clash. The rest of our notation closely follows theirs.

on point estimates does not compound with the sample size. Sampling uncertainty, however, shrinks like $1/\sqrt{T}$, so approximation error matters for inference, classical or Bayesian, once $\sqrt{T}\delta$ is not small. Third, linearisation, which fixes δ, does not fail as the sample grows in the sense FRS claimed, but its bias does grow relative to standard errors. Their likelihood-ratio diagnostic survives: the log likelihood ratio between two approximations differs by $O(T\delta)$, against sampling noise of order $\sqrt{T}$, so the diagnostic detects approximation error in roughly the regime where it matters.

Section 2 locates the error: step (14) of FRS's appendix bounds a difference of two T-fold products of densities by $TB\chi\delta$, but the telescoping identity places $T-1$ density factors beside each per-period error, so B is really the $(T-1)$-th power of a bound on the density. It also gives a counterexample that, unlike AGH's, satisfies FRS's compact-support assumption. Section 3 states and proves Corrected Proposition $2'$. Section 4 draws the consequences for estimation and for Bayesian inference, where FRS's corollaries hold with explicit constants. Following FRS, we state the corrected proposition under high-level assumptions on the conditional densities. They capture three properties: the per-period approximation error in the conditional densities does not grow along the sample, the likelihood forgets the initial state, and the exact and computed stationary distributions are close. As FRS do in their Lemmas 5 and 6, Appendix B derives these assumptions from conditions on the model, now with constants uniform in the sample size; this requires restricting FRS's class to models in which every observable is measured with error and the state transition contracts, in a metric of our choosing, for every realisation of the shocks. This holds globally in the Brock–Mirman model FRS use in their own application and, for small enough shocks, in any determinate DSGE model. A supplementary appendix provides a formalisation of all these results in Lean, verifying their correctness.

2 The setting and the error in the proof

FRS consider the state-space system

$$S_t = \varphi(S_{t-1}, W_t; \gamma), \qquad Y_t = g(S_t, V_t; \gamma),$$

in which S_t is the vector of state variables, both endogenous (such as capital) and exogenous (such as productivity); W_t are the structural shocks that drive the states; Y_t are the observables; and V_t are the shocks to the measurement equation, such as measurement error. The transition equation φ is the equilibrium law of motion of the states, given by the model's policy functions, and the measurement equation g maps the states and measurement shocks into the observables. Neither is known in closed form in general, and the researcher works with numerical approximations φ_j and g_j. Here γ is the vector of structural parameters; we assume, as FRS do, compact supports for the state and the i.i.d. shocks, and a unique invariant distribution μ^* of the state. The shock is partitioned as $W_t = (W_{1t}, W_{2t})$, where (W_{2t}, V_t) is recovered from the data by inverting the system (FRS's Assumption 3). Here T is the number of observations in the data $y^T = (y_1, \ldots, y_T)$, and the likelihoods below are

the exact integrals implied by each model, not simulation estimates of them; the error from evaluating them by simulation, for example with a particle filter, is a separate matter that neither FRS nor this comment addresses.

We fix γ and the data y^T and suppress both from our notation. Write $\omega = W_1^T$ for the unobserved shock path, with law Q, and

$$p_t(s,\omega) = p(y_t \mid W_1^t = \omega_{\le t},\, S_0 = s,\, y^{t-1})$$

for the conditional density of the t-th observation, with $p_{j,t}$ (FRS's $p_j(y_t \mid W_1^t, S_0, y^{t-1})$) its counterpart under the computed equations (φ_j, g_j), whose invariant distribution is μ_j^*. The two likelihoods are[2]

$$L(y^T) = \int_S \int_\Omega \prod_{t=1}^T p_t(s,\omega)\, Q(\mathrm{d}\omega)\, \mu^*(\mathrm{d}s), \qquad L_j(y^T) = \int_S \int_\Omega \prod_{t=1}^T p_{j,t}(s,\omega)\, Q(\mathrm{d}\omega)\, \mu_j^*(\mathrm{d}s).$$

FRS's Assumptions 4 and 8 bound the densities below by $\xi > 0$. Their Lemma 6 gives $|p_{j,t} - p_t| \le \chi\delta$ when $\|\varphi_j - \varphi\| \le \delta$ and $\|g_j - g\| \le \delta$ (it bounds the error in the conditional densities when the numerical errors in the policy function and the measurement equation are bounded). Their Condition 1 is the mean contraction $\int \|\varphi(s,w) - \varphi(s',w)\|\, Q(\mathrm{d}w) \le \alpha\|s - s'\|$ with $\alpha < 1$.

FRS's Proposition 2. *Under Assumptions 1–11 and Condition 1, if $\|\varphi_j - \varphi\| \le \delta$ and $\|g_j - g\| \le \delta$, there are positive constants B and L_F such that, for all T, $|L_j(y^T) - L(y^T)| < (TB\chi + L_F/(1-\alpha))\delta$.*

The proof writes $F(s) = \int \prod_t p_t(s,\cdot)\,\mathrm{d}Q$ and $F_j(s) = \int \prod_t p_{j,t}(s,\cdot)\,\mathrm{d}Q$ and splits

$$|L_j - L| \le \Big|\int F\,\mathrm{d}\mu_j^* - \int F\,\mathrm{d}\mu^*\Big| + \int |F_j - F|\,\mathrm{d}\mu_j^*.$$

The first term is bounded in FRS's (12) by $L_F\delta/(1-\alpha)$, using a result of Santos and Peralta-Alva (2005) with L_F a Lipschitz constant of F. The second is bounded in their (14) by $TB\chi\delta$, from Lemma 6 and a bound B on the densities. Step (14) is where the proof fails. A difference of two products satisfies the telescoping identity

$$\prod_{t=1}^T p_{j,t} - \prod_{t=1}^T p_t = \sum_{t=1}^T \Big(\prod_{s<t} p_{j,s}\Big)\,(p_{j,t} - p_t)\,\Big(\prod_{s>t} p_s\Big), \tag{1}$$

so each of the T per-period errors is multiplied by $T-1$ densities. With M an upper bound on the densities, the bound that follows is $TM^{T-1}\chi\delta$, and M^{T-1} depends on T. No T-free constant can take its place:

Proposition 1. *Let $M > 1$ and $\varepsilon > 0$. There is no constant B such that $|(M+\varepsilon)^T - M^T| \le TB\varepsilon$ for every $T \in \mathbb{N}$.*

[2] FRS write the integral over W_1^T period by period, as a product of integrals. Since p_t depends on the whole history $\omega_{\le t}$, the likelihood is the integral of the product. Nothing below depends on the distinction.

Proof. By Bernoulli's inequality, $(M+\varepsilon)^{n+1}-M^{n+1} = M^{n+1}[(1+\varepsilon/M)^{n+1}-1] \geq (n+1)\varepsilon M^n$. The bound would therefore give $M^n \leq B$ for all n, which is false since $M > 1$. □

Two further steps let constants depend on T without comment. In (12), the Lipschitz constant of F is "different for each t, but since t is finite, we can set a global L" (our L_F), although the proposition is asserted for all T. And Lemma 6 asserts χ for all t "since the model is stationary", although the recovered (v_t, s_t, w_{2t}) depend on a t-fold composition of the transition, whose sensitivity to δ can accumulate with t. The corrected proposition below makes both requirements explicit.

AGH's counterexample uses a Gaussian measurement shock, which does not have the compact support FRS assume. We provide a counterexample with compact support, mainly as an illustration for the reader, but also to show that the failure of Proposition 2 in AGH's counterexample does not hinge on the noncompactness.

Example 2 (A counterexample with compact support)**.** Let f be a smooth density on $[0,1]$ equal to a constant $c_0 > 0$ on $[\frac{1}{4}, \frac{3}{4}]$; a normalised smooth bump function is one, with $c_0 \geq 1$. Take the static model (S_t and W_t trivial) $y_t = g(v_t) = \gamma v_t$ with $v_t \sim f$ and $0 < \gamma < c_0$, approximated by $g_j(v) = (1+\delta)\gamma v$, so that $|g_j(v) - g(v)| \leq \gamma\delta$ on the support. FRS's assumptions hold. For data in $[\frac{1}{4}\gamma(1+\delta), \frac{3}{4}\gamma]$ the exact and computed densities are $M = c_0/\gamma > 1$ and $M/(1+\delta)$, so

$$L(y^T) - L_j(y^T) = M^T(1 - (1+\delta)^{-T}),$$

which for fixed δ grows exponentially in T and eventually exceeds any bound linear in T. With $\gamma > c_0$ instead, $M < 1$ and the difference vanishes exponentially, as in AGH. In both cases $\log L(y^T) - \log L_j(y^T) = T\log(1+\delta)$ exactly.

The last line of the example is the point of the next section: the level of the likelihood moves like M^T, but the log-likelihood gap is linear in T, as FRS's argument suggested.

3 The corrected proposition

The accumulation of error that FRS describe is multiplicative. This was problematic for Proposition 2 in levels, but can be side-stepped by working with the log likelihood, or with the likelihood ratio L_j/L. We prove a corrected Proposition 2′ by working with the log likelihood. We state the result under three assumptions.

Assumption A1 (Per-period error, uniformly in t)**.** There are $\xi > 0$ and $\varepsilon \geq 0$ such that $p_t(s,\omega) \geq \xi$, $p_{j,t}(s,\omega) \geq \xi$ and $|p_{j,t}(s,\omega) - p_t(s,\omega)| \leq \varepsilon$ for all t, s and ω.

In FRS's notation $\varepsilon = \chi\delta$; the lower bounds are their Assumptions 4 and 8.

Assumption A2 (Lipschitz dependence on the initial state)**.** There are $\ell_1, \ldots, \ell_T \geq 0$ such that $\log p_t(s',\omega) - \log p_t(s,\omega) \leq \ell_t\, d(s,s')$ for all t, s, s' and ω. Write $K_T = \sum_{t=1}^{T} \ell_t$.

Only the exact densities need this, not the computed ones.

Assumption A3 (Coupled initial distributions)**.** There is a probability space (X, P) and measurable maps $a, b : X \to S$ with laws μ^* and μ_j^* such that $d(a(x), b(x)) \le \rho$ for P-almost every x.

Assumption A3 replaces FRS's use of Condition 1. It holds with $\rho = \delta/(1-\alpha)$ if Condition 1 is strengthened from contraction in mean to contraction for every shock, $d(\varphi(s,w), \varphi(s',w)) \le \alpha\, d(s,s')$ for all w, and the computed chain's invariant distribution is unique.

Corrected Proposition 2′. *Under Assumptions A1–A3,*

$$e^{-D}\, L(y^T) \;\le\; L_j(y^T) \;\le\; e^{D}\, L(y^T), \qquad D = \frac{T\varepsilon}{\xi} + K_T\, \rho,$$

equivalently $|\log L_j(y^T) - \log L(y^T)| \le D$.

The proof uses two lemmas.

Lemma 3. *If $b \ge \xi > 0$ and $|a - b| \le \varepsilon$, then $a \le e^{\varepsilon/\xi}\, b$. Hence, under Assumption A1, for all s and ω,*

$$\prod_{t=1}^{T} p_{j,t}(s,\omega) \le e^{T\varepsilon/\xi} \prod_{t=1}^{T} p_t(s,\omega) \qquad \text{and} \qquad \prod_{t=1}^{T} p_t(s,\omega) \le e^{T\varepsilon/\xi} \prod_{t=1}^{T} p_{j,t}(s,\omega).$$

Proof. $a \le b + \varepsilon \le b + (\varepsilon/\xi)\, b = (1 + \varepsilon/\xi)\, b \le e^{\varepsilon/\xi}\, b$, using $b/\xi \ge 1$ and $1 + x \le e^x$. The product bounds follow by multiplying T such inequalities between nonnegative numbers; the second uses $p_{j,t} \ge \xi$. □

Lemma 4. *Under Assumption A2, $\prod_{t=1}^{T} p_t(s',\omega) \le e^{K_T d(s,s')} \prod_{t=1}^{T} p_t(s,\omega)$ for all s, s' and ω.*

Proof. Multiply the inequalities $p_t(s',\omega) \le e^{\ell_t d(s,s')}\, p_t(s,\omega)$ over t. □

Proof of Corrected Proposition 2′. Integrate out the shock path:

$$F(s) = \int \prod_t p_t(s,\omega)\, Q(\mathrm{d}\omega), \qquad F_j(s) = \int \prod_t p_{j,t}(s,\omega)\, Q(\mathrm{d}\omega).$$

By Assumption A3 and a change of variables, $L = \int_X F(a(x))\, P(\mathrm{d}x)$ and $L_j = \int_X F_j(b(x))\, P(\mathrm{d}x)$.

Upper bound. For all s, s' and ω, Lemma 3 and then Lemma 4 give

$$\prod_t p_{j,t}(s',\omega) \le e^{T\varepsilon/\xi} \prod_t p_t(s',\omega) \le e^{T\varepsilon/\xi + K_T d(s,s')} \prod_t p_t(s,\omega).$$

Integrating over ω, $F_j(s') \le e^{T\varepsilon/\xi + K_T d(s,s')} F(s)$. Put $s = a(x)$ and $s' = b(x)$. Since $d(a(x), b(x)) \le \rho$ almost surely and $F \ge 0$, $F_j(b(x)) \le e^D F(a(x))$ for P-almost every x, and integrating against P gives $L_j \le e^D L$.

Lower bound. Take the two moves in the other order: first move the initial state within the exact model, then change model. For all s, s' and ω,

$$\prod_t p_t(s,\omega) \le e^{K_T d(s',s)} \prod_t p_t(s',\omega) \le e^{K_T d(s',s)+T\varepsilon/\xi} \prod_t p_{j,t}(s',\omega),$$

by Lemma 4 and the second part of Lemma 3. Integrating over ω and then, with $s = a(x)$ and $s' = b(x)$, over P gives $L \le e^D L_j$. This ordering is why Assumption A2 is needed only for the exact model.

Finally, L and L_j are at least $\xi^T > 0$, since the densities are at least ξ and Q and P are probability measures, so taking logarithms gives the second form. □

Our approach to the proof of Corrected Proposition 2′ differs from FRS's approach to their Proposition 2. To demonstrate that the correction is just about fixing line (14) of their proof, Appendix A gives the same result with the proof laid out step by step as FRS's proof of Proposition 2, so that the two can be read side by side.

Remark 5 (Sharpness). The dependence of D on T cannot be improved. With every exact density equal to M and every computed density equal to $M+\varepsilon$, $\log L_j - \log L = T\log(1+\varepsilon/M)$, which is linear in T; Example 2 is an instance within FRS's framework. In AGH's Gaussian model restricted to a compact support, data with every $y_t = \gamma + c$ give $\log L_j - \log L = T(c\delta - \delta^2/2)/\sigma^2$, again linear in T.

When the per-period Lipschitz constants decay geometrically, as they do when the exact chain forgets its initial state at rate α, the constant K_T is bounded uniformly in T and the bound takes exactly FRS's form, with the level replaced by the log.

Corollary 6. *If $\ell_t \le c\,\alpha^t$ with $0 \le \alpha < 1$ and $\rho = \delta/(1-\alpha)$, then for every T,*

$$\left|\log L_j(y^T) - \log L(y^T)\right| \le \frac{T\varepsilon}{\xi} + \frac{c\,\delta}{(1-\alpha)^2}, \qquad \frac{1}{T}\left|\log L_j(y^T) - \log L(y^T)\right| \le \frac{\varepsilon}{\xi} + \frac{c\,\delta}{T(1-\alpha)^2}.$$

Proof. $K_T \le c\sum_{t\ge 0}\alpha^t = c/(1-\alpha)$; substitute into Corrected Proposition 2′. □

Assumptions A1–A3 are high-level by design, as are the conditions of FRS's Proposition 2 once their Lemmas 5 and 6 are applied. Assumption A1 requires the computed model's per-period density error to be bounded uniformly along the sample, Assumption A2 that the likelihood forget the initial state, and Assumption A3 that the exact and computed stationary distributions be close. FRS derived their high-level assumptions from regularity conditions satisfied by standard macroeconomic models in their Lemmas 5 and 6. Since we modified these high-level assumptions for our Corrected Proposition 2′, we provide updated results in Appendix B showing that standard macroeconomic models satisfy our assumptions: all three hold, with $\varepsilon = \eta + c\delta/(1-\alpha)$, $K_T \le c\alpha/(\xi(1-\alpha))$ and $\rho = \delta/(1-\alpha)$, in models in which every observable is measured with error and the state transition contracts, in a metric of our choosing, for every realisation of the shocks. This holds globally in the Brock–Mirman model of FRS's application and, for small enough shocks, in any determinate DSGE model.

Whether they can be derived in FRS's full class, with only contraction in mean, remains open.

4 Implications

The size of the error. FRS's statement that the error is of order $T\delta$ is correct for the relative error: $|L_j/L - 1| \le e^D - 1$, which is approximately D when D is small. It is false for the absolute error, which inherits the factor M^T carried by the level of the likelihood itself.

Estimation. By Corollary 6, the average log likelihood $\frac{1}{T}\log L_j$ is within $\varepsilon/\xi + O(\delta/T)$ of $\frac{1}{T}\log L$, uniformly in T. This is the dynamic analogue of the level part of AGH's approximation error Δ_j, now derived from the policy-function error and the dynamics of the state rather than assumed. It agrees with AGH's conclusion that approximation error of order δ moves the estimator's probability limit by $O(\delta)$, and not by more as T grows. A bound on the level of the objective alone delivers only $O(\sqrt{\delta})$ for the maximiser under quadratic identification; AGH's $O(\Delta_j)$ rate also uses closeness of the first two derivatives, which our result does not require.

Bayesian inference. FRS's Corollaries 1–4 assert that likelihood ratios, marginal likelihoods, posteriors and posterior expectations of the computed model converge to their exact counterparts. Corrected Proposition 2′ makes these quantitative. Suppose its conclusion holds at every parameter value with a common D, and let $\pi(\cdot \mid y^T) = L/Z$ and $\pi_j(\cdot \mid y^T) = L_j/Z_j$ be the posterior densities with respect to a prior π_0, where $Z = \int L \,\mathrm{d}\pi_0$ and $Z_j = \int L_j \,\mathrm{d}\pi_0$.

Proposition 7. *If $L, L_j > 0$ are π_0-integrable and $Z > 0$, then*

(i) $e^{-D} Z \le Z_j \le e^D Z$;

(ii) $L_j(\gamma')/L_j(\gamma) \le e^{2D}\, L(\gamma')/L(\gamma)$ *for all γ, γ', so every Metropolis–Hastings acceptance ratio is off by at most a factor e^{2D};*

(iii) $e^{-2D}\pi(\gamma \mid y^T) \le \pi_j(\gamma \mid y^T) \le e^{2D}\pi(\gamma \mid y^T)$;

(iv) the total variation distance between the posteriors is at most $(e^{2D} - 1)/2$;

(v) for measurable h with $|h| \le B$, $|\mathbb{E}_j[h \mid y^T] - \mathbb{E}[h \mid y^T]| \le B(e^{2D} - 1)$.

The proof is in Appendix C. Part (v) replaces FRS's Corollary 4, which asserts convergence without a rate, with an explicit bound. The closest result we know is the approximation theorem of Dashti and Stuart (2017, Theorem 4.9 in the arXiv version), building on Cotter et al. (2010): if the negative log likelihoods satisfy $|\Phi - \Phi_N| \le M_2(\|u\|)\psi(N)$, the Hellinger distance between the posteriors is at most $C\psi(N)$ for N large. Corrected Proposition 2′ supplies that hypothesis with $\Phi = -\log L$ and $\psi = \delta$. Proposition 7 is the version with explicit constants, and it holds for every δ.

What remains of FRS's warnings. The claim that linearisation "is due to fail as the sample size grows" because the exact and computed likelihoods diverge does not survive as stated: both levels go to zero or infinity like M^T anyway, and the log-likelihood gap grows no faster than the log likelihood itself. For classical estimation based on the average log likelihood, a fixed δ produces a bias in point estimates that does not grow with T, and the same is true of posterior means in AGH's example. What grows is the bias relative to sampling uncertainty, which shrinks like $1/\sqrt{T}$: AGH's Theorem 4 requires $\sqrt{T}\Delta_j \to 0$ for the approximation to leave the asymptotic distribution unaffected, and in regular models the Bernstein–von Mises theorem gives the same condition for posteriors. The bounds of Proposition 7, which degrade like e^{2D} with $D \propto T\delta$, are conservative by comparison: in AGH's Gaussian example the total variation distance between the exact and computed posteriors is of order $\sqrt{T}\delta/\sigma$, not $T\delta$. Likelihood ratios themselves do move by a factor of order $e^{T\delta}$; in Example 2 the ratio between the exact and the computed likelihood is exactly $(1+\delta)^T$. In corrected form, then, FRS's warning stands with $\sqrt{T}$ in place of T: approximation error matters for inference, classical or Bayesian, once $\sqrt{T}\delta$ is not small, and for comparisons of likelihood levels once $T\delta$ is not small.

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A The proof in FRS's layout

This appendix proves Corrected Proposition 2′ following the structure of FRS's proof of Proposition 2 as closely as possible, with the same decomposition, the same two terms and their equation numbers, primed. In FRS's notation, define for each initial state S_0

$$f_T(S_0) = \int \prod_{t=1}^{T} p_t(S_0, \omega)\, Q(\mathrm{d}\omega), \qquad f_{jT}(S_0) = \int \prod_{t=1}^{T} p_{j,t}(S_0, \omega)\, Q(\mathrm{d}\omega),$$

so that $L(y^T) = \int f_T \,\mathrm{d}\mu^*$ and $L_j(y^T) = \int f_{jT} \,\mathrm{d}\mu_j^*$, as in FRS's proof of Proposition 1. (In Section 3 these are F and F_j.) All four integrals below are at least $\xi^T > 0$, so their logarithms are defined.

The decomposition. FRS split the level difference by adding and subtracting $\int f_T \,\mathrm{d}\mu_j^*$. We split the log difference in the same way:

$$\log L_j - \log L = \underbrace{\left[\log \int f_{jT} \,\mathrm{d}\mu_j^* - \log \int f_T \,\mathrm{d}\mu_j^*\right]}_{\text{model change, distribution } \mu_j^* \text{ fixed}} + \underbrace{\left[\log \int f_T \,\mathrm{d}\mu_j^* - \log \int f_T \,\mathrm{d}\mu^*\right]}_{\text{distribution change, exact model fixed}}. \qquad (2)$$

The second term is the one FRS bound in (12), the first the one they bound in (13)–(14).

Step 1: regularity of f_T in the initial state. FRS show, using their Lemma 5, that f_T is Lipschitz in S_0 with constant L_F, and remark that the constant differs with T. Here the corresponding fact is that $\log f_T$ is Lipschitz with constant K_T: by Lemma 4, $\prod_t p_t(s',\omega) \le e^{K_T d(s,s')} \prod_t p_t(s,\omega)$ for every ω, and integrating over ω,

$$f_T(s') \le e^{K_T d(s,s')}\, f_T(s) \qquad \text{for all } s, s' \in S. \tag{3}$$

The dependence of K_T on T is explicit, and under geometric decay of the ℓ_t (Corollary 6) it is bounded uniformly in T.

Step 2: the distribution term, (12′). FRS bound the second term of their decomposition by $L_F\delta/(1-\alpha)$, applying Theorem 6 of Santos and Peralta-Alva (2005) under Condition 1. We use the coupling of Assumption A3 instead. By change of variables and (3) with $s = a(x)$, $s' = b(x)$, $d(a(x), b(x)) \le \rho$,

$$\int f_T \,\mathrm{d}\mu_j^* = \int_X f_T(b(x))\, P(\mathrm{d}x) \le e^{K_T\rho} \int_X f_T(a(x))\, P(\mathrm{d}x) = e^{K_T\rho} \int f_T \,\mathrm{d}\mu^*,$$

and, exchanging the roles of a and b, $\int f_T \,\mathrm{d}\mu^* \le e^{K_T\rho} \int f_T \,\mathrm{d}\mu_j^*$. Hence

$$\Big|\log \int f_T \,\mathrm{d}\mu_j^* - \log \int f_T \,\mathrm{d}\mu^*\Big| \le K_T\, \rho. \tag{12′}$$

Only the exact model enters this term, which is why Assumption A2 is needed only for the exact densities.

Step 3: the model term, (13′). FRS write the first term of their decomposition as the integral of $f_{jT} - f_T$ against μ_j^*, their (13). On the log scale the first term of (2) is the log of a ratio of two integrals against the same measure,

$$\log \int f_{jT} \,\mathrm{d}\mu_j^* - \log \int f_T \,\mathrm{d}\mu_j^* = \log \frac{\int f_{jT} \,\mathrm{d}\mu_j^*}{\int f_T \,\mathrm{d}\mu_j^*}, \tag{13′}$$

so it is enough to bound f_{jT}/f_T pointwise.

Step 4: bounding the model term, (14′). This is the step that fails in FRS. They apply Lemma 6, $|p_{j,t} - p_t| \le \chi\delta$, to bound $|f_{jT} - f_T|$ by $TB\chi\delta$; as Section 2 shows, the telescoping identity (1) puts $T-1$ density factors beside each $\chi\delta$, so no T-free B exists. Bounding the ratio instead, Lemma 6 in the form of Assumption A1 gives, by Lemma 3,

$$e^{-T\varepsilon/\xi} \prod_t p_t(s,\omega) \le \prod_t p_{j,t}(s,\omega) \le e^{T\varepsilon/\xi} \prod_t p_t(s,\omega)$$

for all s and ω. Integrating over ω gives $e^{-T\varepsilon/\xi} f_T \le f_{jT} \le e^{T\varepsilon/\xi} f_T$ pointwise, then integrating against μ_j^* gives the same bounds for the two integrals in (13′), so

$$\left|\log \int f_{jT}\,\mathrm{d}\mu_j^* - \log \int f_T\,\mathrm{d}\mu_j^*\right| \le \frac{T\varepsilon}{\xi} \;=\; \frac{T\chi\delta}{\xi}. \tag{14′}$$

Step 5: conclusion. Putting together (12′) and (14′) in (2) delivers the result:

$$\left|\log L_j(y^T) - \log L(y^T)\right| \le \frac{T\chi\delta}{\xi} + K_T\rho,$$

which, with $\rho = \delta/(1-\alpha)$ and $K_T \le K$ bounded in T, is $(T\chi/\xi + K/(1-\alpha))\,\delta$, the form of FRS's bound with the level replaced by the log. □

The correspondence with FRS's proof is:

	FRS, Proposition 2	Corrected Proposition 2′
Decomposition	add and subtract $\int f_T\,\mathrm{d}\mu_j^*$	the same, in logs (2)
Regularity in S_0	f_T is L_F-Lipschitz (Lemma 5)	$\log f_T$ is K_T-Lipschitz (3)
Distribution term	(12): $L_F\delta/(1-\alpha)$, Santos–Peralta-Alva	(12′): $K_T\rho$, coupling
Model term	(13)–(14): $TB\chi\delta$, Lemma 6 (fails)	(13′)–(14′): $T\chi\delta/\xi$, Lemma 6
Conclusion	triangle inequality	triangle inequality

B The assumptions from primitives

This appendix is the counterpart, for Corrected Proposition 2′, of FRS's Lemmas 5 and 6 and of their use of Theorem 6 of Santos and Peralta-Alva (2005): it derives Assumptions A1–A3 from regularity conditions satisfied by standard macroeconomic models, with constants uniform in the sample size. Doing so requires three restrictions that FRS do not impose: no W_2, contraction for every shock rather than in mean, and a unique invariant distribution for the computed chain. We work in the case $\dim V_t = \dim Y_t$, so that $W_1 = W$ and the measurement shock alone is recovered from the data. The state path, defined by s_0 given and $s_t = \varphi(s_{t-1}, \omega_t)$, is then a function $s_t(s_0, \omega)$ of the initial state and the shock path only, and $p_t(s_0, \omega) = f_t(s_t(s_0, \omega))$ for a function f_t of the current state: the density of V at $g^{-1}(y_t; s)$ times the Jacobian. Similarly $p_{j,t} = f_{j,t}(s_{j,t})$, with $s_{j,t}$ the computed path. We assume throughout that the state space S is a nonempty compact metric space, that φ and φ_j are jointly measurable and continuous in the state, that φ is a pathwise α-contraction, $d(\varphi(s,w), \varphi(s',w)) \le \alpha\, d(s,s')$ for all w with $0 \le \alpha < 1$, and that $d(\varphi_j(s,w), \varphi(s,w)) \le \delta$ with $\delta \ge 0$.

B.1 The coupling

Lemma 8 (Krylov–Bogolyubov for a random map)**.** *Let X be a nonempty compact metric space, ν a probability measure on a measurable space W, and $F : X \times W \to X$ jointly measurable with $F(\cdot, w)$ continuous for each w. Then there is a probability measure μ on X with $(\mu \otimes \nu) \circ F^{-1} = \mu$.*

Proof. Write $T\mu = (\mu \otimes \nu) \circ F^{-1}$ and $(Mh)(x) = \int h(F(x,w))\,\nu(\mathrm{d}w)$ for bounded continuous h. By change of variables and Fubini's theorem, $\int h\,\mathrm{d}(T\mu) = \int Mh\,\mathrm{d}\mu$. By dominated convergence Mh is continuous, so T is weakly continuous. Start from a Dirac measure μ_0 and form the averages $\bar{\mu}_n = \frac{1}{n+1}\sum_{k=0}^{n} T^k\mu_0$. The difference $\int Mh\,\mathrm{d}\bar{\mu}_n - \int h\,\mathrm{d}\bar{\mu}_n$ telescopes to $\frac{1}{n+1}(\int h\,\mathrm{d}T^{n+1}\mu_0 - \int h\,\mathrm{d}\mu_0)$, which is at most $2\|h\|/(n+1)$ in absolute value. By Prokhorov's theorem the probability measures on X form a compact space, so $(\bar{\mu}_n)$ has a cluster point μ, and by continuity $\int Mh\,\mathrm{d}\mu = \int h\,\mathrm{d}\mu$ for every bounded continuous h. Hence $\int h\,\mathrm{d}(T\mu) = \int h\,\mathrm{d}\mu$ for all such h, and $T\mu = \mu$. □

Theorem 9 (The coupling)**.** *If μ^* is the unique invariant distribution of φ and μ_j^* the unique invariant distribution of φ_j, then Assumption A3 holds with $\rho = \delta/(1-\alpha)$.*

Proof. Let $B_\rho = \{(s,s') \in S \times S : d(s',s) \le \rho\}$, which is closed, hence compact, and contains the diagonal. Define $G((s,s'),w) = (\varphi(s,w), \varphi_j(s',w))$, both chains driven by the same shock. G maps B_ρ into itself, since

$$d(\varphi_j(s',w), \varphi(s,w)) \le d(\varphi_j(s',w), \varphi(s',w)) + d(\varphi(s',w), \varphi(s,w)) \le \delta + \alpha\rho = \rho.$$

It is jointly measurable and continuous in the pair, so Lemma 8 gives an invariant probability measure P on B_ρ. Because the first coordinate of G is φ applied to the first coordinate, taking the first marginal commutes with one step of the chain: $(P_1 \otimes \nu)\circ\varphi^{-1} = ((P \otimes \nu)\circ G^{-1})_1 = P_1$. So P_1 is invariant for φ and equals μ^* by uniqueness; likewise $P_2 = \mu_j^*$. Take $X = B_\rho$ and a, b the coordinate projections. □

Without uniqueness of the computed chain's invariant distribution, the conclusion holds for some invariant distribution of the computed chain, namely the one the coupling selects.

B.2 The densities

Lemma 10. *For all s_0, s_0', ω and t: (a) $d(s_{j,t}(s_0,\omega), s_t(s_0,\omega)) \le \delta/(1-\alpha)$; (b) $d(s_t(s_0,\omega), s_t(s_0',\omega)) \le \alpha^t d(s_0,s_0')$.*

Proof. (a) With $d_t = d(s_{j,t}, s_t)$, the triangle inequality as in the proof of Theorem 9 gives $d_{t+1} \le \alpha d_t + \delta$, so $d_t \le \alpha^t d_0 + \delta/(1-\alpha)$ by induction, and $d_0 = 0$. (b) Induction on t using the contraction. □

Theorem 11 (Corrected Proposition 2′ from primitives)**.** *Suppose each chain has a unique invariant distribution, the measurable functions $f_t, f_{j,t}$ take values in $[\xi, M]$ with $\xi > 0$, $|f_{j,t} - f_t| \le \eta$, and each f_t is c-Lipschitz. Then for every T and every law Q of the shock path,*

$$|\log L_j(y^T) - \log L(y^T)| \le \frac{T}{\xi}\Big(\eta + \frac{c\,\delta}{1-\alpha}\Big) + \frac{c\,\alpha}{\xi}\,\frac{\delta}{(1-\alpha)^2}.$$

Proof. Assumption A1. With $x = s_t(s_0,\omega)$ and $y = s_{j,t}(s_0,\omega)$, $|f_{j,t}(y) - f_t(x)| \le |f_{j,t}(y) - f_t(y)| + |f_t(y) - f_t(x)| \le \eta + c\,d(y,x)$, and $d(y,x) \le \delta/(1-\alpha)$ by Lemma 10(a). So $\varepsilon = \eta + c\delta/(1-\alpha)$, uniformly in t.

Assumption A2. With $x = s_t(s_0, \omega)$ and $x' = s_t(s_0', \omega)$, $|f_t(x') - f_t(x)| \le c\,\alpha^t d(s_0, s_0')$ by Lemma 10(b). Since $f_t(x) \ge \xi$, the first part of Lemma 3 gives $f_t(x') \le e^{(c/\xi)\alpha^t d(s_0,s_0')} f_t(x)$. This is Assumption A2 with $\ell_t = (c/\xi)\alpha^t$, so $K_T \le (c/\xi)\sum_{t\ge1}\alpha^t = c\alpha/(\xi(1-\alpha))$.

Assumption A3 is Theorem 9, with $\rho = \delta/(1-\alpha)$. Corrected Proposition 2′ now gives $D = T\varepsilon/\xi + K_T\rho$, which is at most the stated bound.

Measurability. The state path is jointly measurable in (s_0, ω), by induction on t, so $\prod_t f_t(s_t)$ is jointly measurable and, by Fubini's theorem, its integral over ω is measurable in s_0. The bounds $0 \le \prod_t f_t \le M^T$ give integrability against every probability measure. □

What this leaves informal is the reduction of FRS's measurement equation to the per-state densities f_t, and the bound $\eta = O(\delta)$ on their error at a fixed state, which is FRS's Lemma 6 argument without the time dimension.

B.3 The choice of metric, and local contraction

Remark 12 (The metric is ours to choose). Nothing in this appendix uses the Euclidean distance. The arguments need only a metric d on the state space S such that (S, d) is compact, φ is an α-contraction in d for every shock, $d(\varphi_j(s,w), \varphi(s,w)) \le \delta$, and the densities f_t are c-Lipschitz in d. Two devices make the contraction condition much weaker than it first appears. The first is a change of coordinates, such as logarithms for multiplicative laws of motion. The second is a weighted metric across components of the state. If $s = (x, \zeta)$, where ζ is an exogenous process with $|\zeta' - \tilde\zeta'| \le \rho_z\,|\zeta - \tilde\zeta|$ for every shock, $\rho_z < 1$, and the endogenous component satisfies $|x' - \tilde x'| \le a\,|x - \tilde x| + b\,|\zeta - \tilde\zeta|$ with $a < 1$, then in the metric $d = |x - \tilde x| + \lambda|\zeta - \tilde\zeta|$ with $\lambda > b/(1-\rho_z)$ the transition is a contraction for every shock, with factor $\max(a, \rho_z + b/\lambda) < 1$. The endogenous state need only contract in its own argument; the effect of the exogenous state on it is absorbed by the weight. The price is in the constants: as the factor approaches one, the T-independent term $c\,\alpha\,\delta/(\xi(1-\alpha)^2)$ of Theorem 11 grows.

The next result shows that contraction for every shock holds, in a suitable norm, near any stable steady state when the shocks are small. Write $\|\cdot\|$ for a norm on $\mathbb{R}^n$ and $|\cdot|$ for a norm on the shock space $\mathbb{R}^m$, and $D_s\varphi$ for the Jacobian of φ with respect to the state.

Proposition 13 (Local contraction)**.** *Let $\varphi : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$ be continuously differentiable, with a deterministic steady state $\bar s = \varphi(\bar s, 0)$, and let $\|\cdot\|$ be a norm on $\mathbb{R}^n$ in which the operator norm of $A = D_s\varphi(\bar s, 0)$ satisfies $\|A\| < \alpha$ for some $\alpha < 1$. Then there are $r > 0$ and $\eta > 0$ such that, writing $B = \{s : \|s - \bar s\| \le r\}$, for every shock with $|w| \le \eta$:*

(i) $\|\varphi(s,w) - \varphi(s',w)\| \le \alpha\,\|s - s'\|$ *for all* $s, s' \in B$*;*

(ii) for every $\delta \in [0, (1-\alpha)r/2]$ *and every* φ_j *with* $\|\varphi_j(s,w) - \varphi(s,w)\| \le \delta$, $\varphi_j(s,w) \in B$ *for all* $s \in B$*.*

Hence, if the shocks are supported in $\{|w| \le \eta\}$, the results of this appendix apply with $S = B$ and $d(s,s') = \|s - s'\|$.

Proof. The map $(s, w) \mapsto D_s\varphi(s, w)$ is continuous, so there is $\varepsilon > 0$ such that $\|D_s\varphi(s, w)\| \le \alpha$ and $\|D_w\varphi(s, w)\| \le C$, for some C, whenever $\|s - \bar{s}\| < \varepsilon$ and $|w| < \varepsilon$. Take $r = \varepsilon/2$ and fix w with $|w| \le \eta \le r$. The ball B is convex, so the mean value inequality gives (i). Applied in w along the segment from 0 to w, it also gives $\|\varphi(\bar{s}, w) - \bar{s}\| \le C|w|$. For $s \in B$,

$$\|\varphi_j(s, w) - \bar{s}\| \le \|\varphi_j(s, w) - \varphi(s, w)\| + \|\varphi(s, w) - \varphi(\bar{s}, w)\| + \|\varphi(\bar{s}, w) - \bar{s}\| \le \delta + \alpha r + C|w|,$$

which is at most r when $\delta \le (1 - \alpha)r/2$ and $|w| \le \eta = \min(r, (1 - \alpha)r/(2C))$. This is (ii); taking $\varphi_j = \varphi$ shows that B is invariant under φ too. □

A norm with $\|A\| < 1$ exists whenever the spectral radius of A is below one (Horn and Johnson, 2013, Lemma 5.6.10), and then any α between $\|A\|$ and one will do.

B.4 Benchmark models

Brock–Mirman (global). With log utility, Cobb–Douglas production and full depreciation, the model FRS use in their own application, the policy function is known exactly: $k' = \theta\beta z k^\theta$, with $\theta < 1$ the capital share. Let $\log z' = \rho_z \log z + \epsilon$ with $0 \le \rho_z < 1$ and $|\epsilon| \le \bar{\epsilon}$. In the coordinates $x = \log k$ and $\zeta = \log z$,

$$x' = \log(\theta\beta) + \zeta + \theta x, \qquad \zeta' = \rho_z \zeta + \epsilon.$$

The rectangle $|\zeta| \le \bar{\zeta} = \bar{\epsilon}/(1 - \rho_z)$, $x \in [(\log(\theta\beta) - \bar{\zeta})/(1 - \theta), (\log(\theta\beta) + \bar{\zeta})/(1 - \theta)]$ is compact and invariant, and in the metric of Remark 12, with $a = \theta$, $b = 1$ and any $\lambda > 1/(1 - \rho_z)$, the transition contracts for every shock with factor $\max(\theta, \rho_z + 1/\lambda) < 1$. This holds for any size of shock, with no local argument.

Real business cycle model (local). In the real business cycle model of King et al. (1988), with partial depreciation and CRRA utility, the policy function has no closed form. At the deterministic steady state the Jacobian of the state transition $(k, \zeta) \mapsto (k', \zeta')$ is triangular, with eigenvalues ρ_z and the stable root $\nu_k \in (0, 1)$ of the saddle path, so its spectral radius is below one. The policy depends on the scale σ of the shocks; applying Proposition 13 to $(s, (w, \sigma)) \mapsto \varphi_\sigma(s, w)$, which requires the policy to be continuously differentiable in σ as well, as the perturbation approach presumes, gives contraction for every shock on a ball around the steady state once the shocks are small enough.

New Keynesian models (global or local). In the basic New Keynesian model of Galí (2015, Ch. 3), log-linearised, the only states are the exogenous technology and monetary shocks, autoregressive with coefficients $|\rho_i| < 1$, which contract for every shock in the max-norm with factor $\max_i |\rho_i|$, globally. The log-linear model is solved exactly, so the relevant case is the nonlinear one. With Rotemberg pricing the nonlinear model has no predetermined endogenous variable, so in a Markov equilibrium its states are still the exogenous shocks and the same global argument applies. With Calvo pricing the nonlinear model adds price dispersion as an endogenous state, whose own coefficient at the zero-inflation steady state is the Calvo

parameter $\theta < 1$, so it falls under Proposition 13, as do models with further endogenous states such as interest-rate smoothing, habits or capital. A determinate equilibrium satisfies the Blanchard–Kahn conditions (Blanchard and Kahn, 1980), under which the state transition of the linearised solution has all its eigenvalues inside the unit circle. With the same proviso on the dependence of the policy on the shock scale, these models contract for every shock near the steady state when the shocks are small. The same argument covers the real business cycle model above, and determinate DSGE models generally.

B.5 Beyond this class

When W_{2t} is also recovered from the data, Assumptions A1 and A2 amount to a stability condition on the inverted filter. FRS's Condition 1 does not imply it, and we know of no primitive condition in their setting that does. The closest results, on forgetting of the initial condition in hidden Markov models (Bickel et al., 1998; Douc et al., 2004) and on uniform-in-time approximation of nonlinear filters (Le Gland and Oudjane, 2004), assume a transition density bounded above and below, which a state that moves by a deterministic map of the shocks does not have.

C Proof of Proposition 7

(i) Integrate $L_j \le e^D L$ and $L \le e^D L_j$ against π_0. (ii) $L_j(\gamma') \le e^D L(\gamma')$ and $1/L_j(\gamma) \le e^D/L(\gamma)$. (iii) Divide $L_j(\gamma) \le e^D L(\gamma)$ by $Z_j \ge e^{-D} Z$, and symmetrically. (iv) Write $a = \pi_j(\gamma \mid y^T)$ and $b = \pi(\gamma \mid y^T)$. By (iii), $a \le e^{2D} b$ and $b \le e^{2D} a$, so $a - b \le (e^{2D} - 1)b$ and $b - a \le (1 - e^{-2D})b \le (e^{2D} - 1)b$. Hence $|a - b| \le (e^{2D} - 1)b$, and integrating against π_0, using $\int b \,\mathrm{d}\pi_0 = 1$, gives $\int |\pi_j - \pi| \,\mathrm{d}\pi_0 \le e^{2D} - 1$, which is twice the total variation distance. (v) $|\int h\,(a - b)\,\mathrm{d}\pi_0| \le B \int |a - b| \,\mathrm{d}\pi_0 \le B(e^{2D} - 1)$. □

D Formal verification

Every result above has been verified in Lean 4 with Mathlib, in the repository cited on the title page. The development builds without unproved steps (`sorry`), and every result depends only on Lean's three standard axioms (`propext`, `Classical.choice`, `Quot.sound`). The formal likelihood is $\int_X \int_\Omega \prod_t p_t(a(x), \omega) \,\mathrm{d}Q \,\mathrm{d}P$ for an arbitrary finite set of periods; Theorem 11 converts it to the form against μ^* by change of variables. Paths are relative to `LeanEconomics/`. The formal proof of Corrected Proposition 2′ assembles Lemmas 3 and 4 pointwise before integrating, as in Section 3, rather than through the decomposition of Appendix A; the result proved is the same.

Result	Lean name
Econometrics/ApproxLikelihood	
Proposition 1	`not_exists_linear_bound_prod_sub`
Log ratio linear in T (Remark)	`log_prod_div_prod_const`
Lemma 3	`le_exp_div_mul`, `prod_le_exp_card_mul_prod`
Lemma 4	`prod_le_exp_sum_mul_prod`
Corrected Proposition 2′, the two bounds	`lik_le_exp_mul_lik`, `..._rev`
Corrected Proposition 2′, log form	`abs_log_likelihood_sub_le`
Corollary 6	`abs_log_likelihood_sub_le_of_geometric`
Coupled chain: band, path bound	`dist_step_le_of_le`, `dist_iterate_le`
Distribution/RandomMap	
Lemma 8	`RandomMap.exists_isInvariant`
Econometrics/ApproxLikelihoodModel	
Theorem 9	`exists_coupling`
Lemma 10	`dist_statePath_le`, `dist_statePath_exact_le`
Theorem 11, Assumption A1	`abs_densPath_sub_le`
Theorem 11, Assumption A2	`densPath_le_exp_mul`
Theorem 11	`abs_log_likelihood_sub_le_of_primitives`
Econometrics/LocalContraction	
Remark 12 (weighted metric)	`weighted_contraction`, `weighted_factor_lt_one`
Proposition 13	`local_contraction`
Brock–Mirman: contraction	`brockMirman_contraction`
Brock–Mirman: invariant rectangle	`brockMirman_mapsTo`
Econometrics/AdaptedNorm	
Adapted norm (remark after Proposition 13)	`exists_adapted_norm`
Econometrics/BrockMirmanLikelihood	
Theorem 11 for Brock–Mirman	`brockMirman_abs_log_likelihood_sub_le`
Econometrics/CompactCounterexample	
Example 2: the density f, $c_0 \geq 1$	`bumpDensity_eq_c0`, `one_le_c0`
Example 2: density of γv	`map_scale_withDensity`
Example 2: $L - L_j$ and $\log L - \log L_j$	`lik_sub_lik`, `log_lik_sub_log_lik`
Example 2: no linear bound; $M < 1$	`not_exists_linear_bound_lik_sub`, `tendsto_lik_sub_zero`
Econometrics/NewKeynesian	
NK, Rotemberg: no predetermined variable	`RotembergNK.conds_congr`
NK, Rotemberg: state is the shock vector	`RotembergNK.statePath_eq`
NK, Rotemberg: global contraction	`RotembergNK.next_contraction`
NK, Calvo: dispersion coefficient θ	`dispersionStep_deriv_at_zeroInflation`
Econometrics/ApproxPosterior	
Proposition 7 (i)–(iii)	`evidence_le_exp_mul`, `postDens_le_exp_mul`
Proposition 7 (iv)	`integral_abs_postDens_sub_le`
Proposition 7 (v)	`abs_integral_postDens_sub_le`

Several facts that Appendix B cites rather than proves are formalised in companion LeanEconomics repositories (`DynamicProgramming`, which includes projects on Stokey et al. (1989) and Meyn and Tweedie (2009), `MonetaryEconomics` and `InternationalEconomics`):

- *Spectral radius and norms.* Gelfand's formula for real matrices and its consequence $\|A^k\| \le C\gamma^k$ for every γ above the spectral radius (`exists_norm_pow_le`), and that the associated affine map is an eventual contraction (`exists_isContractionOn_iterate_affineOp`). Building on the first, the adapted norm cited from Horn and Johnson (2013), $\|x\|_* = \sup_k \gamma^{-k}\|A^k x\|$, is formalised here (`exists_adapted_norm`): it is equivalent to the sup norm, and in it $\|Ax\|_* \le \gamma\|x\|_*$ for every γ above the spectral radius.
- *The stable root of the growth model.* For the deterministic one-sector growth model with general concave utility and technology, the real business cycle model without labour supply, the linearised Euler equation has one root in $(0, 1)$ and one above $1/\beta$, and the derivative of the policy function at the steady state is the stable root (`growth_policy_deriv`); also the local stability and stable manifold theorems of Stokey et al. (1989, Theorems 6.5, 6.6) (`theorem_6_5`, `theorem_6_6`).
- *Blanchard–Kahn.* The general theorem for linear rational-expectations systems of any dimension: determinacy holds if and only if there is no unit root, the number of unstable roots equals the number of jump variables, and a genericity condition holds (`determinate_iff_unstableCount`); and in a determinate equilibrium the state transition has all its eigenvalues inside the unit circle (`lam_stable`).
- *Smooth dependence of the solution.* The implicit function theorem gives, for a nonlinear stochastic system $\mathbb{E}_t G(x_{t-1}, x_t, x_{t+1}, e_t) = 0$, a locally unique equilibrium, continuously differentiable in the disturbance process and with derivative equal to the linear rational-expectations solution (`local_determinacyS`). For the class $\mathbb{E}_t f(y_{t+1}, y_t, y_{t-1}, x_t, \sigma\epsilon_{t+1}) = 0$ with an invertible linearisation, the equilibrium is C^k in the shock scale σ and given by a Markov policy (`exists_smooth_policy`), which is the smoothness in σ the real business cycle model of Appendix B uses.
- *Markov equilibrium of the nonlinear New Keynesian model.* The exact Calvo model (the Euler equation, the recursions for the optimal reset price and the exact law of motion of price dispersion) under a Taylor rule satisfying the Taylor principle has, for small enough shocks, a locally unique Markov equilibrium in (Δ_{t-1}, s_t) that is C^2 in the shock amplitude (`markov_equilibrium`, `markov_equilibrium_general`). On a general filtered probability space it is adapted and unique among bounded processes (`seq_existence_ae_adapted`), with a Markov representation for autoregressive shocks with any bounded innovation law (`ar1_markov_representation`). These results assume no government purchases. For the exact Rotemberg model, a Markov equilibrium that is a function of the exogenous state alone exists and is locally unique, for small enough shocks, if and only if the Taylor principle holds in the Rotemberg slope, on every state space and almost surely on filtered probability spaces, with a Markov representation for autoregressive shocks (`NonlinearRotembergMarkov.markov_equilibrium_general`, `NonlinearRotembergAR1.ar1_markov_representation`); it is C^∞ in

the shock scale (`NonlinearRotembergSmooth.markov_equilibrium_smooth`).

- *Brock–Mirman.* The closed-form policy $k' = \theta\beta z k^\theta$ is the unique maximiser of the Bellman equation (`stoch_argmax_unique`).
- *Invariant distributions.* Krylov–Bogolyubov for Feller kernels on compact spaces (`theorem_12_10`), independently of Lemma 8; and uniqueness of the invariant distribution, assumed in Theorem 9, for the stochastic growth model (`growth_invariant_unique`) and under Condition M (`theorem_11_12`).

Theorem 11 is also instantiated for the Brock–Mirman model (`brockMirman_abs_log_likelihood_sub_le`): the state space is the compact invariant rectangle with the weighted metric of Remark 12, $|\Delta \log k| + \lambda |\Delta \log z|$, in which the transition contracts for every shock with factor $\max(\theta, \rho_z + 1/\lambda) < 1$ once $\lambda > 1/(1-\rho_z)$; the computed transition, the densities and the uniqueness of the invariant distributions remain hypotheses, as in the general theorem. What remains unformalised is the existence of a Markov equilibrium of the nonlinear New Keynesian models for shocks that are not small: both the Calvo and the Rotemberg results above are local. For Rotemberg pricing, given a Markov equilibrium, the contraction of Appendix B holds for every state, since the state is the vector of exogenous shocks. For Example 2, the density f is constructed as a normalised smooth bump (`bumpDensity`): smooth, positive exactly on $(0, 1)$, integrating to one and constant on $[\frac{1}{4}, \frac{3}{4}]$. FRS's assumptions are not stated as a formal list, so the claim that the example satisfies them rests on these verified properties of f, g and g_j.